



CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

# Chapter 24: Measures of Central Tendency (Mean, Median, Quartiles and Mode)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

## 1 Mean of raw and tabulated data

1. Assertion (A): The mean of the first 9 natural numbers is 4.5. Reason (R): Mean =  $\frac{\text{sum of all the observations}}{\text{total number of observations}}$ . (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A **ICSE 2025, 1 mark**

**SOLUTION**

$1 + 2 + \dots + 9 = 45$ , so the mean is  $\frac{45}{9} = 5$ , not 4.5. A is false.

R is the correct formula for the mean.

**Answer: (b)**

2. For the data below, find (a) the mean, to the nearest whole number (b) the median.

**ICSE 2025, 3 marks**

| $x$ | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|-----|----|----|----|----|----|----|----|
| $f$ | 3  | 2  | 2  | 6  | 3  | 5  | 3  |

**SOLUTION**

(a)  $\sum f = 24$  and  $\sum fx = 30 + 22 + 24 + 78 + 42 + 75 + 48 = 319$ . Mean =  $\frac{319}{24} \approx 13.29$ , which is 13 to the nearest whole number.

(b) Cumulative frequencies: 3, 5, 7, 13, 16, 21, 24. With  $n = 24$  (even), the median is the mean of the 12th and 13th values.

Values 8 to 13 are all 13, so the 12th and 13th values are both 13.

**Answer: (a) mean = 13 (b) median = 13**

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